

Index

- Absorber, 34, 46, 108, 118, 119, 202, 203
 circuit analog, 29, 185, 202, 203, 276
 Jaumann, 276
Active arrays, 7, 136–213, 274
Admittance, 290–293
 characteristic, 292
 load, 292
Amplifiers, 42, 51, 129, 136, 178, 219, 270, 272, 274
Analytic approach, xxii, 11, 212, 243
Antenna impedance, 18, 23, 25, 26, 27–29, 33, 35, 42, 182–184, 199, 200, 240, 270, 303, 304
Antenna gain, 18, 26, 209
Aperiodicity, 175, 242–268
Appel-Hansen, J., 24
Array gain, 205–210
Array impedance, *See* Impedance, array
Arrays, 24–54, 181–213, 269–271

Backscattering, minimizing, 175–177
Backscattering diagram, 104–114, 145, 148, 161–165, 176
Bahret, W. F., xviii, xxiii
Beth Israel Hospital, Boston, 344

Bilinear transformation, 281–287, 290–293, 344, 345
Bistatic scattering, 6, 12, 31–45, 52–54, 95–103, 113, 116, 133, 139, 142, 166–175, 179, 215–219, 221–223, 225–239, 277
Boeing, 277
Broad band arrays, 181–213, 269–271, 278–280
Broad band matching, 187–200, 211, 274, 275, 288–305

Characteristic impedance, 8, 40–42, 182–184, 189, 190, 193–198, 281–287, 290–305, 331–333, 393
Chu, R. S., 306
Circuit analog absorbers, 29, 185, 202, 203, 276, 277
Circulators, 42, 43, 51
Common misconceptions, 10, 45, 130, 175, 198, 221, 239, 303
Croswell, W. F., xxiii, 269

Damon, E. K., 115
Design philosophy, 198–200, 205, 212

- Dielectric, 10, 11, 13, 49, 77, 78, 84, 85, 133, 134, 181–186, 188–198, 205, 212, 223, 252, 253, 270–275, 306, 310, 315–326
- Dominek, A., xxiv
- Edge effect, 138–165, 271, 272
- Edge waves, 5, 11–13
- Education, engineering, xxi, xxii, 222, 277, 278, 288
- Ehrenspeck, H. W., 115
- Element impedance, 24
- Element types
 - four legged loaded, 259–261, 268
 - gangbuster, 138, 185, 221, 269
 - Jerusalem cross, 253
 - snowflake, 254, 255
 - snowflake, modified, 254–256
 - tripoles, 249–252, 254, 256–258
- Embedded impedance, 327–345
- End current, 5, 94, 97, 98
- Engineering education, xxii, 222, 278, 288
- English, E. K., xxiii, 261, 268
- Epis, J. J., 306
- ESP program, 333, 336, 337, 340, 341
- Equivalent transmission line, 183–186, 189–198
- Evanescent waves, 27, 29, 30, 84, 87–91, 93, 130, 145, 184, 199, 273
- Exotic materials, 304
- Feed organizer, 278, 279
- Finite x infinite array, 1–14, 29, 30, 56–83, 84–135, 136–180, 182, 271–274, 275, 276, 327–345
- Floquet's theorem, 2–6, 64, 77, 85, 86, 90, 117, 132, 143, 178, 310, 329, 330
- Foster's reactance theorem, 201, 202, 248, 304
- Future directions, 276, 278
- Garbacz, R. J., xxiv, 23, 24, 115
- Gibson, J., xxiv
- Graphical approaches, 281–287, 289–300
- Grating lobes, 12, 27, 36, 39, 46, 47, 75, 82, 85, 88, 90, 105–107, 110, 117, 124, 126, 133, 182, 205, 206, 246, 248, 249, 270, 273, 276
- diagram, 88
- trapped, 5, 12, 273
- Green, R. B., 23
- Groundplane
 - effect upon bandwidth, 211
 - finite x infinite, 137–141, 143–176
 - infinite x infinite, 184–186, 199–202
 - impedance, *See* Impedance
 - magnetic, 209, 210
 - moving, 200–202
- Hansen, R. C., 24
- Hansen, W. W., 11, 115, 209
- Hansen-Woodyard condition, 11, 115, 209
- Harris Corporation, xxiii, 186, 269, 278–280
- Heidrich, E., 24
- Henderson, L. W., 1
- Hexagon element, 252, 253
- Hill, R. A., 246, 247, 249
- Horn scattering, 226–229, 265–267
- Hughes, J. A., 217
- Hybrid periodic structures, 16, 324
- Hybrids, 31, 33, 40–42, 44, 45, 50, 54, 216, 217, 221, 222, 232–234, 238
- Hewlett Packard, 289
- Imaginary space, 86, 87, 117, 125, 127
- Impedance
 - antenna, 17, 18, 23, 25, 26, 29, 33, 35, 42, 183–206, 240, 270, 302–305
 - array, 211, 270, 271
 - characteristic, 28, 40, 184, 189, 190, 192, 194, 196, 197, 281–287, 289–305, 330
 - components, 184, 186–198, 211, 270
 - discussion, 74, 75, 343, 344
 - element, 24
 - embedded, 327–345
 - generator, 7–10
 - groundplane, 183, 187–190, 193–195, 199–202, 211, 270
 - input, 40–42, 173, 184, 188–192, 194, 200, 219, 281–287, 290–296, 308–326
 - intrinsic, 182, 188, 270, 316
 - load, 3, 4, 8–13, 17–29, 31–42, 49, 52–55, 76, 77, 140–179, 194,

215–222, 238, 240, 244, 267,
268, 281–304, 330–345
matching, 186, 277, 288–305
measurements, 340–343
mutual, 52–54, 61–82, 105, 141, 178,
184, 244–249, 272–276, 301,
328–345
approach, 76, 85, 86, 329
overcoupled, 332–345
scan, 7–9, 12, 13, 38, 85–89, 92, 93,
105–107, 125–130, 137,
192–198, 271, 272, 324,
327–345
self, 76, 202, 204, 244–249, 301,
330–345
stick, 57–83, 85–87, 89–115,
141–146, 275, 276, 327–345
terminal, 56, 175–179, 183, 187–191,
205, 233, 278, 305
Induced voltage 61–64
Infinite x infinite array, 1–5, 24–39, 63,
75, 82, 83, 85, 181–205, 275,
328–330
Input impedance, See Impedance, input
Inter-element spacings, 1–5, 12, 13, 27,
76, 82, 83, 85–87, 98, 117, 134,
183, 184, 187, 206, 211, 244, 245,
247–249, 252, 256, 270, 273, 276,
301, 330
Intrinsic impedance, See Impedance,
intrinsic
Invisible space, 86, 88, 91, 92

Janning, D. S., xxiii, 82, 117, 127–130,
276
Anomaly, 128
Jaumann absorber, 276
Jerusalem cross, 252, 253
Jitter, 146, 159, 161–165, 178, 179, 272
Johnson, B. L., 145 (reference 75)

Kennaugh, E. M., 23, 24
Kent, B. M. xxiii
King, D. D., 24
Kornbau, T. W., 185, 269
Kouyoumjian, R. G., xxiii
Kraus, J. D., xxiii, 209, 205

Larson, C. J., 45
Lee, K. M., 306
Leeper, W. J., 261
Lerner, D. S., 306
Lin, S. J., 278

Load impedance, see Impedance, load
Logan, J., xxiii

Marino, R. A., 306 Magnetic
groundplane, 209, 210
Mailloux, R. J., 115
Marhefka, R. J., 205, 209
Matching tools, 289–293
general discussion, 303, 304
tricks, 296–304
McEntee, J. A., 23
Meanderlines, 306–326, 275
Mehr, J., 241
Meloling, J., xxiii
Minimizing backscattering, 175–177
Minimum scattering antenna (MSA)
definition, classical, 34–51
definition, relaxed, 35
examples, 32, 35–40, 50
Micromanagement of backscattered field,
140–146
Mission Research Corporation, xxiii,
186, 268, 280
Mode suppressors, 256–260
Multiband designs, 252–258, 275
Munk, B. A., xi, xvii, 117
Munk, P., xxiii, 77, 117
Mutual impedance, See Impedance,
mutual
Mutual impedance approach, 76, 85, 86,
329

Near field, 29, 30

Omnidirectional antenna with low RCS,
214–223
Optimization, 10, 11, 175–177, 193,
212, 213, 243, 271, 276–278, 296,
326
Overcompensation, 297–299

Pakan, J. J., 306
Parabolic antennas (2-D), 224–241
Pattern function or pattern factor, 31–34,
46, 47, 61–63, 74, 81, 246, 275
Periodic moment method (PMM), xxiii,
1–5
program, 1–5, 12, 24, 27, 75, 81, 84,
133, 182, 190–193, 242–244,
249, 273, 330, 331
Peters, L., Jr., xviii, xix, xxiii, 23, 24

- Plane of incidence, definition, 1–4, 53, 245
- Plane wave expansion, 77–81, 83, 85, 90–93, 246–249
- Poisson's sum formula, 132
- Polarization factor, 17, 19, 20
- Polarizer, meanderline, 275, 306–326
- Position circles, 281–287, 293–297
- Pryor, J. B., xxiii, 117, 296, 326, 344, 345

- Radar cross section,
 - antennas, 15–23, 49–56, 222, 271, 272
 - arrays, 24–29, 31–34, 36, 37, 45–49, 160, 165, 177–179, 181, 271–277
 - cancellation, 22
 - definition, 19
 - design approach, 22, 23, 48
 - flat plate, 55
 - general, xvii, 7, 12, 49–51, 51, 52, 55, 56, 85, 117, 123, 134, 136, 137, 180, 181, 222, 272–274, 304
 - history, 23, 24
 - horns, 46
 - omnidirectional, 214–223, 272
 - parabolic, 214, 224–241, 272
 - residual mode, 17, 20, 21, 49, 50, 178, 271
 - “small” antennas, 48, 49
 - tapered aperture, 37–40, 43–45
 - Yagis, 52–54
- Radomes, 15, 16, 117–123, 238, 239
- Reflection coefficient, 18–22, 24–27, 28, 29, 50–52, 56, 77, 135, 137, 140, 143–146, 158–164, 172–175, 181, 246, 249, 272, 282–287, 290–293, 303, 304, 307, 308, 326, 340–343
- Richmond, J. H., 115
- Ripples, 3–10, 12, 13, 29, 30
- Rockway, J., xxiii
- Rohde and Schwartz, 289
- Rumsey, V. H., 289

- Saladin, E. C., 85 (reference 74)
- Scan compensation, 191, 192, 197, 198, 322–326
- Scan impedance, *See* Impedance, scan
- Scattering diagram, 104–107, 110, 148, 161–165, 176
- Scattering from faceted radomes, 117–125
- Scattering pattern, *see* Bistatic scattering
- Schneider, S. W., xxiii
- Self impedance, *See* Impedance, self
- Simon, J. I., 219, 220, 221, 223
- Single stub tuner, 293, 294
- Smith chart, 21–23, 24–27, 281–287, 289–302, 307–324, 331–341
- Snell's law, 316
- Spatial approach, 56–82
- Spectral approach, 56–82
- SPLAT program, 1–10, 12, 24, 75, 85, 93, 94, 128–134, 138–140, 161, 182, 224, 225, 333–339
- Stealth, 46, 140
- Stick arrays, 56–85, 86–93
- Structural scattering, 45, 46, *see also* Residual scattering
- Surface waves,
 - active arrays, 136–180
 - calculation, 93, 94
 - control of, 7–10, 108–117, 136, 147–165, 178, 179, 199
 - effect of, 3–7, 12, 13, 30, 56, 85, 136, 178, 179
 - excitation, 90–93
 - forced, 84, 86, 87, 89, 90
 - free, 84, 86, 87, 89, 90
 - from corners, 117–123
 - from discontinuities, 123–125
 - from groundplane, 129, 130, 147–151
 - model for studying, 146, 147
 - passive arrays, 84–135
 - previous work, 115, 116
 - Type I, definition and discussion, 5, 11–13, 84, 85, 133, 134, 272–274
 - Type II, definition and discussion, 4, 5, 11–13, 84, 85, 133, 134, 272–274
 - radiation from, 6, 7, 11, 85, 94–103, 114–116, 133
 - variation with angle of incidence, 97–100, 111–114
 - variation with array size, 100–102
 - variation with frequency, 102–107, 110, 163–165, 176
 - when scanning E-plane, 123–129

- Tapered aperture, 15, 37–40, 43–45
- Tapered periodic surfaces, 261–267

- Taylor, R. C., xxiii, 269, 278
Terminal impedance, *See* Impedance,
 terminal
Terret, C., 306
Theory, 56–82, 275, 276
Thevenin equivalent circuit, 7–10, 32,
 35–38, 48, 50
Thomanek, L., 289
Transformation circles, 281–287,
 293–299
Triad, definition, 141–142
 radiation, 142–146, 162–176, 178,
 179
Tuning, staggered, 242
- Ufimtsev, P. Y., 12
Usoff, J. M., 1, 75, 82, 85, 134, 276
Utt, E. L., xxiii
- Wang, J. J. H., 24
Weber, E., 185, 204
Wheeler, H. A., 184
Wiesbeck, W., 24
Woodyard, J. R., 11, 115, 209
- Yagi-Uda, 115
Young, L., 306